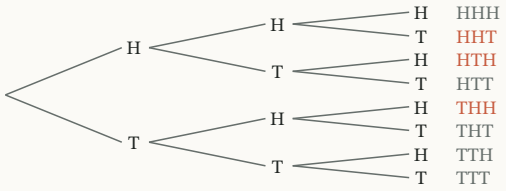
Probability CHEAT SHEET

EXPERIMENTS AND SAMPLE SPACES

coin: $S = \{H, T\}$, $n(S) = 2$
die: $S = \{1, 2, 3, 4, 5, 6\}$, $n(S) = 6$
two coins: HH, HT, TH, TT ($HT \neq TH$)
three coins: $2 \times 2 \times 2 = 8$ (a tree)
3 drinks, 2 snacks: $3 \times 2 = 6$ pairs
coin, then die: H1 … H6, T1 … T6: 12
heads: roll a die; tails: toss again: 8
only the total of two dice: $S = \{2, \ldots, 12\}$



three coins: 8 tips · exactly two heads: 3 of 8

TWO DICE: 36 OUTCOMES

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

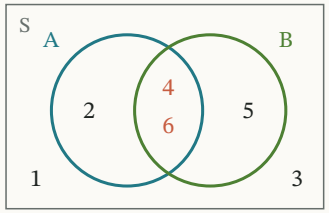
36 equally likely ordered pairs: $(3, 4) \neq (4, 3)$

sevens: the longest diagonal, 6 cells

EVENTS

event: a subset of S $E = \{2, 4, 6\}$ (even)
simple: one outcome $\{4\}$ compound: $\{2, 4, 6\}$
sure: $E = S$ impossible: $E = \varnothing$
two dice, total 7: 6 cells on one diagonal

NOT, AND, OR (A EVEN, B MORE THAN 3)



not A = A′ = {1, 3, 5}
A or B = A ∪ B = {2, 4, 5, 6}
A and B = A ∩ B = {4, 6}
A but not B = A ∩ B′ = {2}

MUTUALLY EXCLUSIVE AND EXHAUSTIVE

mutually exclusive: $A \cap B = \varnothing$ (can't both happen)
exhaustive: $A \cup B \cup C = S$ (one must happen)
both: a partition $\{1, 2\}, \{3, 4\}, \{5, 6\}$

WORDS INTO SETS

two coins, at least one head = {HH, HT, TH}
no head = {TT} = not (at least one head)
exactly one head {HT, TH} and no head:
  mutually exclusive, not exhaustive (HH left out)
cards 1–10: prime {2, 3, 5, 7}, even {2, 4, 6, 8, 10}
  prime and even = {2} prime or even: 8 numbers

THE CLASSICAL DEFINITION

$P(E) = \frac{n(E)}{n(S)}$ (equally likely outcomes)

$P(\text{even}) = \frac{3}{6} = \frac{1}{2}$ $P(5 \text{ or more}) = \frac{2}{6} = \frac{1}{3}$

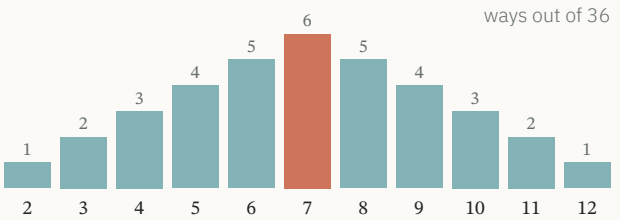
$0 \le P(E) \le 1$ $P(7) = 0$ $P(\text{less than } 7) = 1$

3 red, 5 blue, 2 green: $P(\text{blue}) = \frac{5}{10} = \frac{1}{2}$

MISSISSIPPI: $P(S) = \frac{4}{11}$ $P(P) = \frac{2}{11}$

½ = 0.5 = 50% ¼ = 0.25 = 25%

TOTALS OF TWO DICE



$P(7) = \frac{6}{36} = \frac{1}{6}$ the most likely total

$P(\text{double}) = \frac{6}{36}$ $P(\text{even total}) = \frac{18}{36} = \frac{1}{2}$

differ by 2: $\frac{8}{36} = \frac{2}{9}$ not $\frac{1}{11}$ per total

THE LONG RUN

relative frequency = $\frac{\text{times it happened}}{\text{number of trials}}$

sevens in 10, 100, 1000, 10 000 rolls (simulated):

0, 0.13, 0.146, 0.17 → $\frac{1}{6} \approx 0.167$

Probability CHEAT SHEET

THE AXIOMS OF PROBABILITY

1. $P(E) \geq 0$ for every event E
2. $P(S) = 1$
3. $A \cap B = \varnothing \Rightarrow P(A \cup B) = P(A) + P(B)$

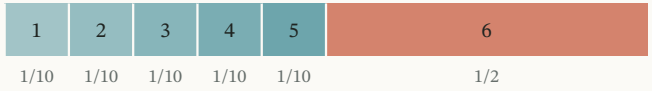
so: $P(\varnothing) = 0$ $\quad 0 \leq P(E) \leq 1$ $\quad$ E inside F $\Rightarrow P(E) \leq P(F)$

outcomes $\omega_1 \ldots \omega_n$: each $P(\omega_i) \geq 0$, they add to 1

$P(E)$ = the sum of $P(\omega)$ over the outcomes in E

n equally likely: each $\frac{1}{n} \Rightarrow P(E) = \frac{n(E)}{n(S)}$

THE PAINT PICTURE



one pot of paint, total 1, spread over S

never negative · separate pieces add

PROBABILITY MODELS

loaded die: 1–5 at $\frac{1}{10}$, six at $\frac{1}{2}$: total 1, valid

$P(\text{even}) = \frac{1}{10} + \frac{1}{10} + \frac{1}{2} = \frac{7}{10}$

0.5, 0.3, 0.2 valid $\quad$ 0.6, 0.5, −0.1 no $\quad$ 0.4, 0.4, 0.4 no

$0.4 + 0.25 + 2p = 1 \Rightarrow p = 0.175$

$P(k)$ proportional to k: $P(k) = \frac{k}{21}$ $\quad P(\text{even}) = \frac{12}{21} = \frac{4}{7}$

THE ADDITION RULE

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

even or more than 3: $\frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{2}{3}$

mutually exclusive: $P(A \cup B) = P(A) + P(B)$

two dice, 7 or 11: $\frac{6}{36} + \frac{2}{36} = \frac{2}{9}$

1 to 20, multiple of 3 or 5: $\frac{6 + 4 - 1}{20} = \frac{9}{20}$

THE COMPLEMENT RULE

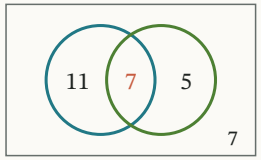
$P(\text{not A}) = 1 - P(A)$

$P(\text{not a six}) = \frac{5}{6}$ $\quad$ rain 70% ⇒ no rain 30%

at least one six (two dice) $= 1 - \frac{25}{36} = \frac{11}{36}$

at least one …: count the opposite, none

VENN REGIONS



a class of 30:
18 football, 12 basketball,
7 both
either: 23/30
exactly one: 16/30

A but not B: $P(A) - P(A \cap B)$

neither: $1 - P(A \cup B)$

exactly one: $P(A) + P(B) - 2P(A \cap B)$

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cup B) = 0.7 \Rightarrow P(A \cap B) = 0.2$

0.7 and 0.6 can't be exclusive: overlap ≥ 0.3

COUNTING

m ways, then n ways: $m \times n$ $\quad$ PIN: $10^4 = 10\,000$

no repeated digit: $10 \times 9 \times 8 \times 7 = 5040$ (0.504)

$n! = n(n-1) \ldots 2 \cdot 1$ $\quad 4! = 24$ $\quad 0! = 1$

${}^nP_r = \frac{n!}{(n-r)!}$ $\quad {}^8P_3 = 8 \times 7 \times 6 = 336$

${}^nC_r = (\text{n choose r}) = \frac{n!}{r!\,(n-r)!} = \frac{{}^nP_r}{r!}$

${}^5C_2 = {}^5C_3 = 10$ $\quad {}^{10}C_3 = 120$ $\quad {}^{52}C_2 = 1326$

Ana and Ben side by side (4 in a row): $\frac{3! \times 2}{4!} = \frac{1}{2}$

A DECK OF CARDS

52 cards: 4 suits × 13 ranks; 12 face cards

$P(\text{king}) = \frac{4}{52} = \frac{1}{13}$ $\quad P(\text{heart}) = \frac{1}{4}$

$P(\text{heart or king}) = \frac{13 + 4 - 1}{52} = \frac{4}{13}$

two aces: $\frac{{}^4C_2}{{}^{52}C_2} = \frac{6}{1326} = \frac{1}{221}$ $\quad$ two hearts: $\frac{1}{17}$

five hearts: $\frac{{}^{13}C_5}{{}^{52}C_5} = \frac{1287}{2\,598\,960} \approx 1$ in 2000

COMMITTEES AND BIRTHDAYS

6 girls, 4 boys, team of 3: ${}^{10}C_3 = 120$ teams

all girls: $\frac{{}^6C_3}{120} = \frac{1}{6}$ $\quad$ 2 girls, 1 boy: $\frac{{}^6C_2 \times {}^4C_1}{120} = \frac{1}{2}$

at least one boy: $1 - \frac{1}{6} = \frac{5}{6}$ $\quad (20 + 60 + 36 + 4 = 120)$

birthdays, 23 people (365 equal days):

$P(\text{a shared birthday}) = 1 - \frac{{}^{365}P_{23}}{365^{23}} \approx 0.5073$